User Study Plan: Evaluating Matrix Factorization Movie Recommender

Hypothesis:

The matrix factorization-based recommender system provides more relevant and satisfying movie recommendations for new users (cold start) than a popularity-based baseline.

Study Design:

- Between-subjects design: Each participant interacts with either the matrix factorization system or a popularity-based system.
- Both systems have identical UI and rating workflow.

Recruitment:

- Target: 40 participants (20 per group), diverse in age, gender, and movie-watching habits.
- Recruitment via university mailing lists, social media, and flyers.
- Incentive: Movie ticket raffle or small gift card.

Procedure:

1. Informed consent and demographic survey.
2. Each participant rates 5 randomly selected movies.
3. System generates recommendations (matrix factorization or popularity-based).
4. Participant reviews top 5 recommendations and rates their relevance (1-5 scale).
5. Participant answers short questionnaire on satisfaction, perceived novelty, and trust.

Data Collected:

- Demographics
- Ratings given to initial movies
- Relevance ratings for recommendations
- Satisfaction, novelty, and trust scores
- Qualitative feedback (optional)

Analysis:

- Compare mean relevance and satisfaction scores between groups (t-test or Mann-Whitney U)
- Analyze effect of demographics on outcomes
- Thematic analysis of qualitative feedback

Ethics:

- Obtain IRB/ethics approval
- Ensure informed consent and data privacy

Reporting:

- Summarize findings, limitations, and implications for recommender system design